

Test 1 · Geometry

Grade 8

40 minutes 100 marks 11 questions No calculator

Lessons 1–6 · Revision of Grade 7 · Polygons · Parallelogram: properties and tests

- Draw a diagram for every question — an unlabelled answer earns no method marks.
- Give a reason for each step: the angle fact, the property, or the test you used.

NAME

CLASS

DATE

MARK / 100

A

Section A · Six marks each

EASY30 marks

One fact each. A short reason is enough.

- 1

Two angles of a triangle are 38° and 57° . Find the third angle. [6]
- 2

An isosceles triangle has a base angle of 72° . Find its apex angle. [6]
- 3

Each exterior angle of a regular polygon is 40° . How many sides has it? [6]
- 4

In parallelogram $ABCD$, $\angle B = 115^\circ$. Find $\angle A$ and $\angle C$. [6]
- 5

In quadrilateral $ABCD$, $AB = CD$ and $AB \parallel CD$. Is $ABCD$ a parallelogram? Name the test you used. [6]

B

Section B · Ten marks each

MEDIUM40 marks

Show the working. Method marks are available even when the final answer is wrong.

- 6

In $\triangle ABC$, $AB = AC$ and $\angle A = 40^\circ$. The bisector of $\angle B$ meets AC at D . Find $\angle BDC$. [10]
- 7

The interior angles of a convex polygon add to 1440° . Find the number of sides, and the size of each interior angle if the polygon is regular. [10]
- 8

The perimeter of a parallelogram is 54 cm and one side is 5 cm shorter than the other. Find the two sides. [10]

- 9 In quadrilateral $ABCD$ the diagonals meet at O , with $AO = OC = 6$ cm, $BO = 5$ cm and $OD = 5$ cm. Is $ABCD$ a parallelogram? Give your reason. [10]

C Section C · Fifteen marks each

HARD

30 marks

Set your work out as a proof: statement, then reason.

- 10 A convex polygon has interior angles adding to 1800° .
- (a) Find the number of sides. [5]
 - (b) The polygon is regular. Find each interior angle and each exterior angle. [5]
 - (c) Find the total number of diagonals. [5]

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- 11 $ABCD$ is a quadrilateral in which $AB = CD$ and $BC = AD$.
- (a) Prove that $ABCD$ is a parallelogram. [10]
 - (b) A student says “ $AC = BD$ would also prove it”. Give a counter-example. [5]